

code.r

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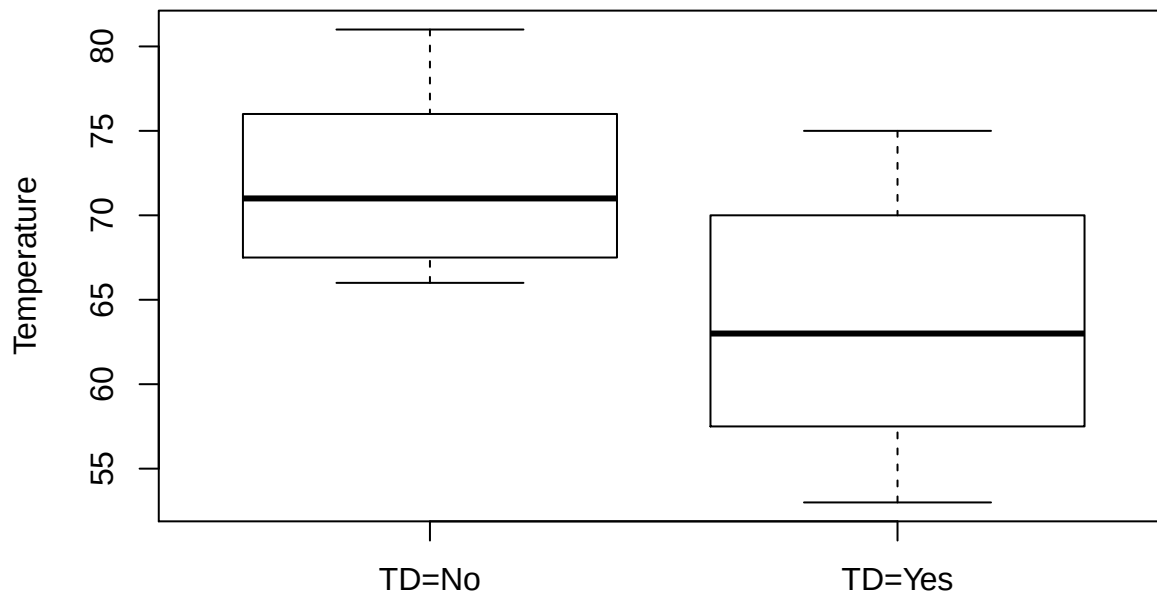
Sun Sep 30 13:07:30 2018

```
setwd("~/Dropbox/Classes/536/CaseStudies/Challenger")

#this is the inverse logit function
invlogit <- function(x)
{
  exp(x)/(1+exp(x))
}

#read the data
#This dataset gives information about the 23 space shuttle flights #before the Challenger disaster. We
#of the flight and whether at least one primary O-ring suffered
#thermal distress
data <- read.table('data.txt',header=TRUE);
attach(data);

#preliminary examination of the data
boxplot(Temp~TD,names=c("TD=No","TD=Yes"),ylab="Temperature")
```



```
#fit a logistic regression model
mylogit <- glm(TD~Temp,family=binomial(link=logit))

#examine the coefficients of the regression
summary(mylogit)

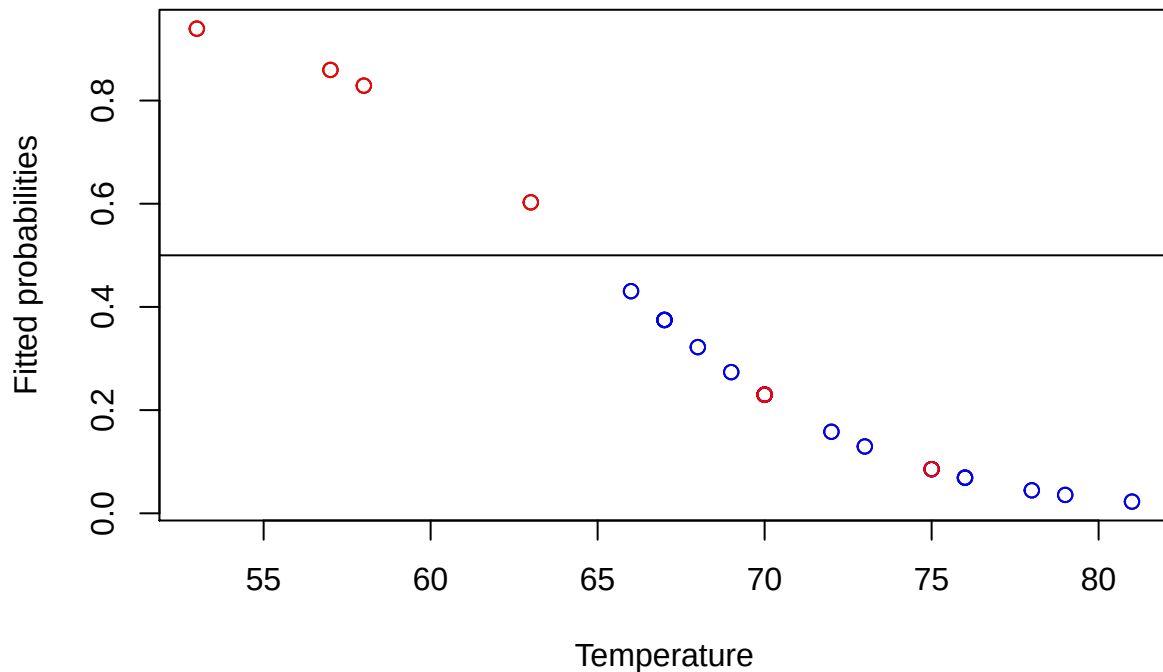
##
## Call:
## glm(formula = TD ~ Temp, family = binomial(link = logit))
##
## Deviance Residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -1.0611 -0.7613 -0.3783  0.4524  2.2175
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  15.0429     7.3786   2.039  0.0415 *
## Temp        -0.2322     0.1082  -2.145  0.0320 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 28.267  on 22  degrees of freedom
## Residual deviance: 20.315  on 21  degrees of freedom
## AIC: 24.315
##
## Number of Fisher Scoring iterations: 5
```

```
#determine which summaries are available
names(mylogit)
```

```
## [1] "coefficients"      "residuals"         "fitted.values"
## [4] "effects"           "R"                  "rank"
## [7] "qr"                "family"             "linear.predictors"
## [10] "deviance"          "aic"                 "null.deviance"
## [13] "iter"              "weights"             "prior.weights"
## [16] "df.residual"       "df.null"             "y"
## [19] "converged"         "boundary"            "model"
## [22] "call"              "formula"             "terms"
## [25] "data"              "offset"              "control"
## [28] "method"            "contrasts"           "xlevels"
```

```
#plot the fitted values
#first plot everything
plot(Temp,mylogit$fitted.values,xlab="Temperature",ylab="Fitted probabilities")
#plot only the "no thermal distress" observations in blue
points(Temp[TD==0],mylogit$fitted.values[TD==0],col="blue")
#plot only the "thermal distress" observations in red
points(Temp[TD==1],mylogit$fitted.values[TD==1],col="red")
abline(h=0.5)
```



```
#the function "fitted(mylogit)" also gives you the fitted values
```

```
#how good is the fit of the model?
```

```
#calculate the Brier Score
```

```
(brier = sum((TD-mylogit$fitted.values)^2))
```

```
## [1] 3.186114
```

```
#calculate the error rate = proportion of incorrectly predicted samples
```

```
(error.rate <- mean((mylogit$fitted.values>0.5 & TD==0) | (mylogit$fitted.values<0.5 & TD==1)))
```

```
## [1] 0.1304348
```

```
#what about Challenger? The temperature at its launch was 31F
```

```
#calculate the predicted probability of thermal distress
```

```
(challenger = invlogit(coef(mylogit)%*%c(1,31)))
```

```
## [1,]
```

```
## [1,] 0.9996088
```

```
#how could they ever have decided to launch the shuttle that day?
```

```
#now let's do the prediction with a more evolved function
```

```
predict(mylogit,data.frame(Temp=31),type='response')
```

```
## [1]
```

```
## 0.9996088
```

```
#determine the standardized residuals
```

```
(myres = (TD-mylogit$fitted.values)/sqrt(mylogit$fitted.values*(1-mylogit$fitted.values)))
```

```
##      1      2      3      4      5      6
## -0.8694280  1.8298705 -0.6137523 -0.6892974 -0.7741411 -0.4332640
##      7      8      9     10     11     12
## -0.3857795 -0.5464867  0.4046177  0.8119436  1.8298705 -0.2159094
##     13     14     15     16     17     18
```

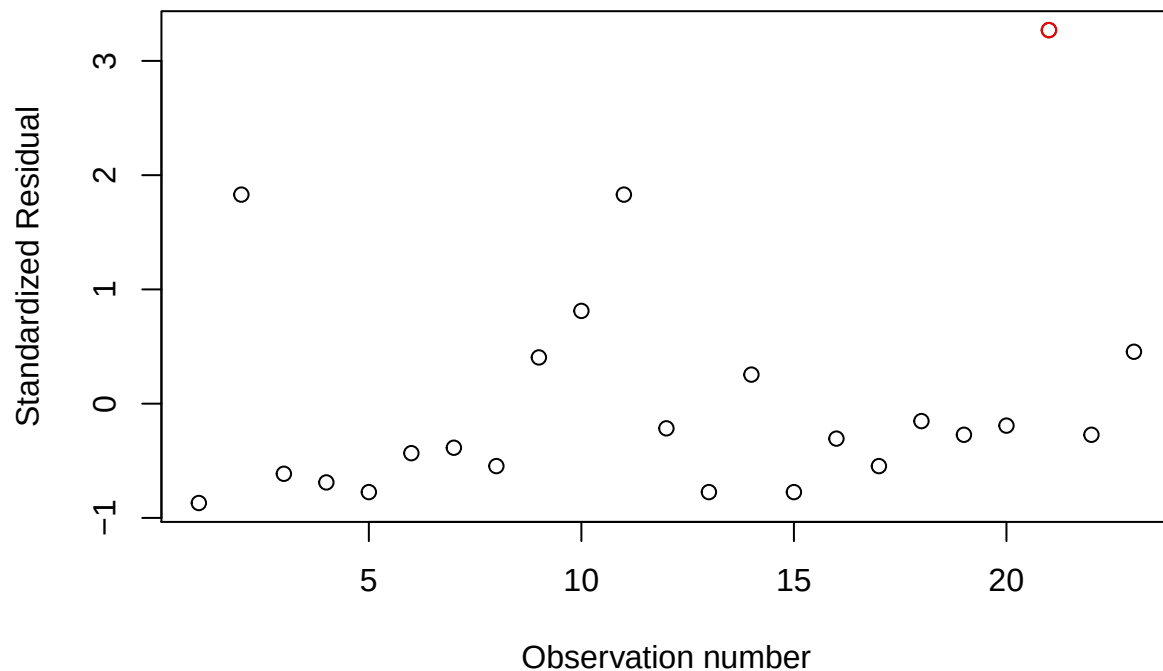
```
## -0.7741411  0.2543261 -0.7741411 -0.3058526 -0.5464867 -0.1524162
##          19          20          21          22          23
## -0.2723320 -0.1922464  3.2695491 -0.2723320  0.4544210
```

```
#calculate the p-value for the chisq test
(1-pchisq(sum(myres^2),length(TD)-2))
```

```
## [1] 0.3350299
```

```
#make an index plot of standardized residuals against observation number
plot(Ft,myres,xlab="Observation number",ylab="Standardized Residual")
```

```
#the outliers are outside the (-2,2) interval
a = Ft[abs(myres)>=2]
points(a,myres[a],col='red')
```



```
#calculate the aic with the formula
mylogit$deviance+2*mylogit$rank
```

```
## [1] 24.31519
```

```
#you can also get it directly
mylogit$aic
```

```
## [1] 24.31519
```

```
#calculate the bic
(bic = mylogit$deviance+mylogit$rank*log(length(TD)))
```

```
## [1] 26.58618
```

```
BIC(mylogit)
```

```
## [1] 26.58618
```

```
#####
#fit two other logistic regressions and compare them
```

```

#this is the model with no repressors
mylogit1 <- glm(TD~1,family=binomial(link=logit))
#compute the Brier score, error rate, aic, bic
(brier1 = sum((TD-mylogit1$fitted.values)^2))

## [1] 4.869565

(error.rate1 <- mean((mylogit1$fitted.values>0.5 & TD==0) | (mylogit1$fitted.values<0.5 & TD==1)))

## [1] 0.3043478
mylogit1$aic

## [1] 30.26715
AIC(mylogit1)

## [1] 30.26715
(bic1 = mylogit1$deviance+mylogit1$rank*log(length(TD)))

## [1] 31.40265
BIC(mylogit1)

## [1] 31.40265
#now let's include a quadratic term
Temp2 = Temp^2
mylogit2 <- glm(TD~Temp+Temp2,family=binomial(link=logit))
#compute the Brier score, error rate, aic, bic
(brier2 = sum((TD-mylogit2$fitted.values)^2))

## [1] 2.987337

(error.rate2 <- mean((mylogit2$fitted.values>0.5 & TD==0) | (mylogit2$fitted.values<0.5 & TD==1)))

## [1] 0.1304348
mylogit2$aic

## [1] 25.3887
AIC(mylogit2)

## [1] 25.3887
(bic2 = mylogit2$deviance+mylogit2$rank*log(length(TD)))

## [1] 28.79519
BIC(mylogit2)

## [1] 28.79519
#test to see if we can delete temperature from the first model
1-pchisq(mylogit1$deviance-mylogit$deviance,1)

## [1] 0.004803533
#test to see if we can delete temperature^2 from the third model
1-pchisq(mylogit$deviance-mylogit2$deviance,1)

```

```
## [1] 0.3357766
```